# SIQUR

Optimal Camera Placement, 3D Spatial UI, and Crowdsourced Forensic Intelligence

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## The Problem

Over 1 billion CCTV cameras are deployed worldwide, yet placement relies on installer intuition rather than mathematical optimization. The result: predictable blind spots, wasted hardware budgets, and zero feedback from post-incident review back into system design. SIQUR closes this loop with three integrated components: **(1)** mathematically optimal camera placement via combinatorial optimization, **(2)** a photorealistic 3D spatial interface built on Gaussian splatting for immersive tactical awareness, and **(3)** OpenClaw, a crowdsourced forensic annotation protocol that converts community investigation into ML training data.

## Camera Placement Model

We formalize camera placement as a 3D variant of the *Art Gallery Problem* (Chvátal, 1975), which proves that $\lfloor n/3 \rfloor$ guards always suffice to cover a simple polygon with $n$ vertices. For a building polytope $\mathcal{P} \subset \mathbb{R}^3$, we discretize the interior into a coverage grid $G = \{g_1, \ldots, g_n\}$ and define candidate camera positions $C = \{c_1, \ldots, c_m\}$ on wall surfaces at mounting height $h = 2.5\,\text{m}$. Each camera $c_i$ has pose parameters $\boldsymbol{\theta}_i = (\phi, \psi, \alpha, r)$ representing yaw, pitch, field-of-view angle, and effective range. The **visibility function** is defined as:

$$v(c_i, g_j) = \begin{cases} 1 & \|c_i - g_j\| \le r \;\wedge\; \angle(c_i, g_j) \le \frac{\alpha}{2} \;\wedge\; \text{clear}(c_i, g_j) \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

where $\text{clear}(c_i, g_j)$ denotes an unoccluded line of sight, verified via raycasting against $\mathcal{P}$. The placement objective reduces to a **Maximum Coverage Location Problem**, equivalent to weighted Set Cover and therefore NP-hard (Feige, 1998):

$$\max_{S \subseteq C,\, |S| \le k} \quad \mathcal{C}(S) = \sum_{j=1}^{n} w_j \cdot \mathbb{1}\Big[\exists\, c_i \in S : v(c_i, g_j) = 1\Big] \tag{2}$$

where $w_j$ is the priority weight of grid point $j$ (elevated near building entries, high-traffic corridors, and asset zones), and $k$ is the user-specified camera budget. The standard greedy algorithm achieves an approximation ratio of $(1 - 1/e) \approx 0.632$. We augment this with a **genetic algorithm** (Yildiz et al., 2014) operating on binary chromosomes $\mathbf{x} \in \{0,1\}^m$ with fitness function:

$$f(\mathbf{x}) = \frac{\sum_j w_j \cdot \mathbb{1}\big[\exists\, i : x_i = 1 \;\wedge\; v(c_i, g_j) = 1\big]}{\sum_j w_j} - \lambda \max\big(0,\, \|\mathbf{x}\|_1 - k\big) \tag{3}$$

with tournament selection, single-point crossover ($p_c = 0.8$), bit-flip mutation ($p_m = 0.02$), converging within ∼200 generations. The output coverage ratio $\rho \in [0,1]$ is surfaced to the user as a placement quality metric.

## 3D Spatial Interface

Building environments are captured via photogrammetry and represented as **3D Gaussian Splats** (Kerbl et al., SIGGRAPH 2023). Each scene primitive $\mathcal{G}_k$ is parameterized by position $\boldsymbol{\mu}_k$, covariance $\boldsymbol{\Sigma}_k$, opacity $\alpha_k$, and spherical harmonic color $\mathbf{c}_k$. The rendered pixel color is computed via differentiable alpha compositing:

$$C(\mathbf{p}) = \sum_{k=1}^{K} \mathbf{c}_k\, \alpha_k \prod_{j=1}^{k-1} (1 - \alpha_j) \cdot \exp\Big(-\tfrac{1}{2} \Delta\mathbf{p}_k^\top \boldsymbol{\Sigma}_k'^{-1} \Delta\mathbf{p}_k\Big) \tag{4}$$

where primed quantities denote 2D screen-space projections. This enables real-time rendering at >30 fps with sub-centimeter visual fidelity, allowing operators to navigate any placed camera's exact viewpoint. A Mapbox GL satellite layer provides global context, with cinematic iris-wipe transitions from orbital zoom to building interior.

## OpenClaw: Crowdsourced Forensics

After an incident, relevant camera feeds are exposed through OpenClaw, a structured crowdsourced investigation interface. Community participants annotate temporal segments, tag persons of interest, and flag anomalous events. Each annotation is a tuple $(t, \mathbf{b}, \ell, p_u)$: timestamp, bounding box, semantic label, and annotator confidence. Following the dual-use computation paradigm (von Ahn et al., 2008), consensus labels are computed as:

$$\hat{\ell}_t = \arg\max_{\ell} \sum_{u \in U_t} p_u \cdot \mathbb{1}[\ell_u = \ell] \tag{5}$$

High-agreement annotations feed a continual learning pipeline for anomaly detection models, creating a **flywheel**: more incidents analyzed → better training data → stronger models → fewer blind spots → faster resolution.

## Architecture

| Component | Technology |
|---|---|
| Placement Engine | OR-Tools + GA, Python, FastAPI |
| 3D Scene Rendering | Gaussian Splatting, Three.js |
| Satellite Globe | Mapbox GL |
| Frontend | Next.js, Framer Motion |
| Video Simulation | Qwen 2.5, NVIDIA DGX |

**References**

[1] Chvátal, *J. Comb. Theory B*, 1975. [2] Feige, *J. ACM*, 1998. [3] Kerbl et al., *ACM TOG*, 2023. [4] Yildiz et al., *IEEE Trans. Comp.*, 2014. [5] von Ahn et al., *Science*, 2008.